21. Name the chemicals used in gene transfer.
22. What do you know about the word pBR332?
23. Mention the application of Biotechnology.
24. What are restriction enzyme. Mention their type with role in Biotechnology.
25. Is their any possibilities to transfer a suitable desirable gene to host plant without vector? Justify your answer.
26. How will you identify a vectors?
27. Compare the various types of Blotting techniques.
28. Write the advantages of herbicide tolerant crops.
29. Write the advantages and disadvantages of Bt cotton.
30. What is bioremediation? give some examples of bioremediation.
31. Write the benefits and risk of Genetically Modified Foods.

Glossary

3' Hydroxy end: The hydroxyl group attached to 3' carbon atom of sugar of the terminal nucleotide of a nucleic acid.

Bacterial artificial chromosomes (BAC): A cloning vector for isolation of genomic DNA constructed on the basis of F-factor.

Chimeric DNA: A recombinant DNA molecule containing unrelated genes.

Cleave: To break phosphodiester bonds of dsDNA, usually with a restriction enzyme.

Cloning site: A location on a cloning vector into which DNA can be inserted.

Cloning: Incorporation of a DNA molecule into a chromosomal site or a cloning vector.

Cloning Vector: A small, self-replicating DNA inserted in a cloning gene.

COS sites: The 12-base, single strand, complementary extension of phage lambda (l) DNA.

DNA Polymerase: An enzyme that catalyses the phosphodiester bond in the formation of DNA.

Endonucleases: An enzyme that catalyses the cleavage of DNA at internal position, cutting DNA at specific sites.

Genome: The entire complement of genetic material of an organism.

Insert DNA: A DNA molecule incorporated into a cloning vector.

Ligase: An enzyme used in genetic engineering experiment to join the cut ends of dsDNA.

M-13: AssDNA bacteriophage used as vector for DNA sequencing.

Phagemid: A cloning vector that contains components derived from both phage DNA and plasmid.

Plasmid: Extrachromosomal, self-replicating, circular dsDNA containing some non-essential genes.

Restriction map: A linear array of sites on DNA cleaved by various restriction enzymes.

Shuttle Vector: A plasmid cloning vector that can replicate in two different organisms due to the presence of two different origin of replication Ori^{EUK} and $Ori^{E.\ coli}$

Taq polymerase: A heat stable DNA polymerase isolated from a thermophilic bacterium *Thermus aquaticus*.

Vectors: Vehicles for transferring DNA from one cell to another.

Biofuel: Fuels like hydrogen, ethanol and methanol produced from a biological source by the action of microorganisms.

Bioleaching: Process of using microorganisms to recover metals from their ores or contaminant

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